

Fourfold forrelation: a self-contained proof

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Abstract

Four binary phase masks encode a scalar that a two-pass interferometer can read directly. Three photons suffice to determine its promised sign with error below one third, at total dose six. We prove that a single parallel interrogation needs dose at least $(2/15)N^{1/8}$ for powers of two $N \geq 2^{30}$. The argument proceeds from the optical experiment to parity functions, then constructs inputs whose small correlations defeat low-dose measurements. All mathematical ingredients are proved here. At $N = 4096$ we also give an explicit finite criterion; the estimate needed to turn that reduction into a dose-six lower bound remains open.

1 The question and the answer

Let $N = 2^d$. Index modes by the binary vectors in $\mathbb{F}_2^d$ and put

$$H_{ij} = N^{-1/2}(-1)^{i \cdot j}, \quad u = N^{-1/2}(1, \dots, 1)^\top.$$

The dot product in the exponent is modulo two. Thus H is real, symmetric and orthogonal: $H^2 = I$. Each mask is a sign vector $x^{(b)} \in \{\pm 1\}^N$, represented by $D_b = \text{diag}(x^{(b)})$. The quantity of interest is

$$F(x) = u^\top D_1 H D_2 H D_3 H D_4 u. \tag{1}$$

It lies in $[-1, 1]$, since every matrix in the product is orthogonal. For any vector v we also write $D_v = \text{diag}(v)$. The input is promised to satisfy either $F \geq 1/4$ or $F \leq -1/4$. The task is to decide which, with error at most $1/3$ on every promised input.

What is counted. A photon crossing one mask costs one unit of dose. A photon crossing two masks costs two. A *hard* dose bound holds on the entire support of the signal state, including every branch of a superposition; it is not an average. Arbitrary reference systems, called idlers, may bypass the masks without cost.

In a *parallel interrogation*, the whole input state is prepared before any mask output is available. Each signal photon crosses the direct-sum mask $D_1 \oplus \dots \oplus D_4$ once, and an arbitrary joint measurement follows. Entanglement, repeated occupation of a mode, and coherence between photon numbers are allowed. Jointly prepared time bins are included; feeding an earlier output into the preparation of a later probe is a different model.

Theorem 1. *For the promise problem above:*

1. *For every N , three photons making two mask crossings each have hard dose six and worst-case error at most $81/256 < 1/3$.*
2. *For every power of two $N \geq 2^{30}$, a parallel interrogation with worst-case error at most $1/3$ has hard dose*

$$D \geq \frac{2}{15}N^{1/8}.$$

The second bound exceeds six for powers of two $N \geq 2^{44}$. It does not settle $N = 4096$. Section 8 develops that finite question separately and identifies the precise remaining estimate. The main proof uses elementary linear algebra and probability; the Gaussian tools used along the way are proved in Appendix A.

2 Why two passes are enough

The product in (1) can be split into the inner product of two unit vectors:

$$L_x = D_2 H D_1 u, \quad R_x = H D_3 H D_4 u, \quad \langle L_x, R_x \rangle = F(x).$$

Prepare one photon in a uniform mode state and an equal superposition of two paths, $(|0\rangle + |1\rangle)u/\sqrt{2}$. Along path zero apply D_1, H, D_2 ; along path one apply D_4, H, D_3, H . Both paths cross exactly two masks. The state becomes

$$\frac{|0\rangle L_x + |1\rangle R_x}{\sqrt{2}}.$$

Recombine the paths and measure in the basis $(|0\rangle \pm |1\rangle)/\sqrt{2}$, discarding the mode label. The output port $X \in \{\pm 1\}$ satisfies

$$\Pr(X = \pm 1 \mid x) = \frac{1 \pm F(x)}{2}.$$

On either promise, this flag is correct with probability at least $5/8$. Repeat independently three times and take the majority. The probability of at least two wrong flags is at most

$$(3/8)^3 + 3(5/8)(3/8)^2 = 81/256 < 1/3.$$

Every run uses dose six, and no outcome is discarded. This proves the first part of Theorem 1.

3 The lower-bound strategy

A lower bound must limit *every* allowed preparation and measurement. We do this by finding two random input laws, μ_+ and μ_- , that are hard to distinguish. They have two complementary properties:

- Each law places more than $9/10$ of its mass on its own promise. Thus any correct solver must accept noticeably more often under μ_+ .
- Products of a few input signs have almost the same expectation under the two laws. A low-dose parallel experiment cannot collect enough of these small differences to distinguish them.

The second point needs more than a bound on polynomial degree: there may be many small correlations. The next section bounds their combined effect on any measurement. Sections 5–7 then construct the laws, prove concentration, and obtain the contradiction $1/5 < 1/12$.

4 What a parallel measurement can learn

4.1 From photons to parity functions

Write $m = 4N$ for the total number of signal coordinates. An occupation vector $\nu = (\nu_1, \dots, \nu_m)$ with $n = \sum_j \nu_j \leq D$ embeds as the normalized symmetric sum of length- D words containing ν_j copies of j and $D - n$ copies of 0. Distinct occupations give orthogonal sums, so this is an isometry, including photon-number superpositions. The extra symbol 0 means an unused *signal* slot; all bypass modes belong to the unrestricted idler. Additional labels on which the mask acts trivially, such as time-bin labels, can likewise be retained in a slot's idler factor and then grouped with the other idlers. The query acts as

$$O_x^{\otimes D}, \quad O_x|0\rangle = |0\rangle, \quad O_x|j\rangle = x_j|j\rangle \quad (1 \leq j \leq m).$$

Allowing arbitrary slot states, beyond symmetric ones, only enlarges the class of experiments against which we prove the bound.

For a word a , let ∂a be the set of nonzero labels occurring oddly. Repeated occurrences cancel in pairs because $x_j^2 = 1$. The word's phase is therefore the parity function $\chi_{\partial a}(x)$, where $\chi_S(x) = \prod_{j \in S} x_j$. Purify the input and write

$$|\psi\rangle = \sum_a |a\rangle \otimes |\psi_a\rangle, \quad \sum_a \|\psi_a\|^2 = 1.$$

The idler vectors ψ_a need not be orthogonal. Let $f(x)$ be the probability of answering *positive*. A measurement effect $0 \leq E \leq I$ gives this probability. Center it by putting $T = E - I/2$, so $\|T\| \leq 1/2$. Then

$$f(x) - \frac{1}{2} = \sum_{a,b} \chi_{\partial a \Delta \partial b}(x) \langle \psi_a, T_{ab} \psi_b \rangle, \quad T_{ab} = (\langle a| \otimes I) T (|b\rangle \otimes I). \quad (2)$$

Here Δ is symmetric difference: labels common to both sets cancel. The parity functions form an orthonormal basis for functions on sign strings under the uniform measure. Consequently

$$f(x) = \sum_S \hat{f}(S) \chi_S(x), \quad \hat{f}(S) = 2^{-m} \sum_x f(x) \chi_S(x),$$

and (2) shows $\hat{f}(S) = 0$ when $|S| > 2D$.

4.2 A bound on an entire Fourier level

We use one elementary matrix fact. If a scalar matrix factors as $K_{ab} = \langle p_a, q_b \rangle$, with feature norms at most P, Q , then for any block operator T ,

$$\|[K_{ab} T_{ab}]\| \leq PQ \|T\|. \quad (3)$$

Attach q_b to input block b , apply T without changing the attached factor, and contract output block a against p_a . The three operator norms are at most $Q, \|T\|, P$. This also proves the claim for arbitrary idler spaces. Here and below an unadorned norm is the Euclidean vector norm or its induced operator norm.

Lemma 2 (Fourier collection bound). *For $1 \leq r \leq \min(2D, m)$ and complex c_S ,*

$$\left| \sum_{|S|=r} c_S \hat{f}(S) \right| \leq \frac{1}{2} \binom{2D}{r} m^{r/4} \max_{|S|=r} |c_S|. \quad (4)$$

Proof. We partition the pairs (a, b) in (2) with $|\partial a \triangle \partial b| = r$. Each differing label occurs oddly on exactly one side. Mark its *last* occurrence on that side. This assigns a unique record of r marked positions to the pair.

For example, $a = (1, 1, 2, 3)$ and $b = (2, 4, 0, 0)$ have parity sets $\{2, 3\}$ and $\{2, 4\}$. Mark the 3 in a and the 4 in b . Removing these marks leaves the same parity set $\{2\}$ on both sides. The repeated 1 and the unused slots contribute no parity.

Fix a record with k marks in a and $r - k$ in b . A word is valid on its own side if the marked labels are nonzero, distinct, occur oddly, and are marked at their last occurrences. Write A, B for the marked sets and $R_a = \partial a \setminus A$, $R_b = \partial b \setminus B$ for the residual sets. The pair belongs to the record exactly when both words are valid, $A \cap B = \emptyset$, and $R_a = R_b$. Indeed, then $\partial a = R_a \dot{\cup} A$ and $\partial b = R_a \dot{\cup} B$, so precisely the marked labels differ.

Index an auxiliary matrix C by k -element sets A and $(r - k)$ -element sets B , putting $C_{AB} = c_{A \cup B}$ if they are disjoint and zero otherwise. Factor it using standard basis vectors on its shorter side. For example, if there are fewer rows, use a basis vector for each row and the corresponding column of C for each column. The product of the largest feature norms is at most

$$\sqrt{\min \left\{ \binom{m}{k}, \binom{m}{r-k} \right\} \max |c_S|} \leq m^{\min(k, r-k)/2} \max |c_S|.$$

Attach to each feature the basis vector labelled by its residual set, and set the feature to zero for an invalid word. Their inner product now enforces residual equality and validity, giving the *exact* record matrix K , with the same norm bound. By (3), the absolute value of its contribution $\langle \psi, [K_{ab} T_{ab}] \psi \rangle$ is at most half that bound. There are $\binom{D}{k} \binom{D}{r-k}$ records with this split. Hence

$$\left| \sum_{|S|=r} c_S \hat{f}(S) \right| \leq \frac{1}{2} \max |c_S| \sum_{k=0}^r \binom{D}{k} \binom{D}{r-k} m^{\min(k, r-k)/2}.$$

Use $\min(k, r - k) \leq r/2$ and count all ways to choose r positions among $2D$ slots to obtain (4). $\square$

The lemma converts small input moments into a bound on any experiment. If two laws have moment differences $\delta_r = \max_{|S|=r} |(\mathbb{E}_+ - \mathbb{E}_-) \chi_S|$, then

$$|\mathbb{E}_+ f - \mathbb{E}_- f| \leq \frac{1}{2} \sum_{r=1}^{\min(2D, m)} \binom{2D}{r} m^{r/4} \delta_r. \quad (5)$$

The constant Fourier coefficient cancels because both laws have mass one.

5 Inputs with small correlations

We want four sign blocks with appreciable forrelation but very small low-order moments. Three independent correlated links supply the forrelation. Random block signs make every distinguishing moment use all four blocks.

Fix $\rho = 200/201$. Independently for $s = 1, 2, 3$, draw jointly Gaussian standard vectors Z_s, W_s with $\mathbb{E} Z_s W_s^\top = \rho H$, and set $a_s = \text{sgn } Z_s$, $b_s = \text{sgn } W_s$ componentwise. This covariance is valid because $\|\rho H\| < 1$. Join the links by componentwise multiplication, denoted $\odot$:

$$y = (a_1, b_1 \odot a_2, b_2 \odot a_3, b_3).$$

To define μ_σ for $\sigma \in \{\pm 1\}$, choose independent uniform signs $\zeta_1, \zeta_2, \zeta_3$, set $\zeta_4 = \sigma \zeta_1 \zeta_2 \zeta_3$, and put $x^{(b)} = \zeta_b y^{(b)}$. Since F is linear in each block, $F(x) = \sigma F(y)$.

5.1 A single link

For coordinate sets A, B write $\mathcal{L}(A, B) = \mathbb{E} \prod_{i \in A} a_i \prod_{j \in B} b_j$. Let $a = |A|$, $b = |B|$, and suppose $a + b \leq r \leq \sqrt{N}$. Then

$$|\mathcal{L}(A, B)| \leq \left(\frac{r}{2\sqrt{N}} \right)^{\max(a, b)}. \quad (6)$$

If just one set is empty, its moment is zero because the other marginal has independent symmetric signs; if both are empty, the moment is one. Otherwise the cross-covariance $R = \rho H_{A, B}$ has

$$\|R\| \leq \left(\sum_{i, j} |R_{ij}|^2 \right)^{1/2} = \rho \sqrt{ab/N} \leq r/(2\sqrt{N}) \leq 1/2.$$

A product of signs in a independent Gaussian coordinates has no Hermite terms of degree below a : it is odd in each coordinate. Here Hermite expansion means orthogonal polynomial expansion for Gaussian inputs. The same holds with b on the other side. The Gaussian contraction proved in Appendix A pairs only equal degrees p and contracts degree p by at most $\|R\|^p$. Both sign products have L^2 norm one. Their common degrees start at $\max(a, b)$, proving (6).

5.2 Three links and four odd blocks

For $S = (S_1, \dots, S_4)$ put $q_b = |S_b|$ and $r = \sum_b q_b$. Averaging the block signs gives

$$\mathbb{E}_\zeta \prod_{b=1}^4 \zeta_b^{q_b} = \sigma^{q_4} \prod_{j=1}^3 \mathbf{1}_{\{q_j \equiv q_4 \pmod{2}\}}.$$

The difference between the two laws therefore vanishes unless all four q_b are odd. In that case independence of the links gives

$$(\mathbb{E}_{\mu_+} - \mathbb{E}_{\mu_-}) \chi_S = 2\mathcal{L}(S_1, S_2)\mathcal{L}(S_2, S_3)\mathcal{L}(S_3, S_4).$$

The three exponents in (6) have sum at least $3r/4$:

$$\begin{aligned} \max(q_1, q_2) &\geq (3q_1 + q_2)/4, \\ \max(q_2, q_3) &\geq (q_2 + q_3)/2, \\ \max(q_3, q_4) &\geq (q_3 + 3q_4)/4. \end{aligned}$$

Because the base in (6) is at most one, we conclude

$$\delta_r \leq 2^{1-3r/4} r^{3r/4} N^{-3r/8} \quad (r \leq \sqrt{N}), \quad \delta_r = 0 \quad \text{unless } r \geq 4 \text{ is even.} \quad (7)$$

The scale of the final result is already visible: at degree four the moment is $O(N^{-3/2})$, while the Fourier bound costs $O(D^4 N)$. Their product is $O(D^4/\sqrt{N})$, which becomes constant at $D = N^{1/8}$.

6 Why these inputs satisfy the promise

Small moments alone do not make hard inputs: the laws must also place most of their mass on the required promises. We prove this by computing the mean and bounding the variance.

6.1 The mean is beyond the threshold

The Gaussian sign identity proved in Appendix A gives

$$\mathbb{E}a_i b_j = \beta_N H_{ij}, \quad \beta_N = \frac{2\sqrt{N}}{\pi} \arcsin(\rho/\sqrt{N}).$$

Expanding (1) and using the three independent links,

$$\mathbb{E}_{\mu_\sigma} F = \frac{\sigma \beta_N^3}{N} \sum_{i,j,k,l} H_{ij}^2 H_{jk}^2 H_{kl}^2 = \sigma \beta_N^3.$$

The sum has N^4 terms, each equal to N^{-3} . Since $\arcsin t \geq t$ for $t \geq 0$ and $\pi < 22/7$,

$$\beta_N^3 - \frac{1}{4} > \left(\frac{1400}{2211}\right)^3 - \frac{1}{4} > \frac{1}{260}. \quad (8)$$

The final comparison is the integer inequality $130 \cdot 1400^3 - 33 \cdot 2211^3 = 38\,842\,277 > 0$.

6.2 Separate independent noise from smooth variation

Lemma 3. *Under either law, $\text{Var } F < 1540/N$.*

Proof. Realize each link using independent standard Gaussian vectors G, ξ, η as

$$a = \text{sgn}(G + \xi/\sqrt{200}), \quad b = \text{sgn}(HG + \eta/\sqrt{200}).$$

After standardizing the marginal variances, the cross-covariance is ρH , as required. Condition on $\mathcal{G} = (G_1, G_2, G_3)$. All primitive signs are then independent, and each is used in only one coordinate of y . Hence the coordinates of y are conditionally independent too. Their primitive means are $\phi(G)$ and $\phi(HG)$, where

$$\phi(t) = 2\Phi(\sqrt{200}t) - 1 = \text{erf}(10t), \quad L_\phi = \|\phi'\|_\infty = 20/\sqrt{\pi}, \quad L_\phi^2 < 128.$$

Here Φ is the standard normal distribution function; the last bound uses $\pi > 25/8$. Since $F(x) = \sigma F(y)$, the gauges do not affect its variance. We use the two terms of total variance:

$$\text{Var } F = \mathbb{E} \text{Var}(F \mid \mathcal{G}) + \text{Var } \bar{F}, \quad \bar{F} = \mathbb{E}(F \mid \mathcal{G}). \quad (9)$$

First term: independent sign noise. For independent signs, variance tensorization bounds the variance of a multilinear function by the sum of its expected squared coefficients in each sign; Appendix A gives the short proof. Each endpoint block of F contributes exactly $1/N$ to this sum: its other chain has norm one, and the endpoint vector is flat. For block two the coefficient of coordinate j is $\ell_j v_j$, where

$$\ell = HD_{a_1} u, \quad v = HD_{b_2 \odot a_3} HD_{b_3} u.$$

These vectors depend on disjoint links and are independent. The coordinates of a_1 are independent uniform signs, so $\mathbb{E} \ell_j^2 = 1/N$; also $\|v\| = 1$. Averaging the sum gives $1/N$. Block three has the same bound by reversing the argument, using the independent uniform signs in b_3 . Consequently

$$\mathbb{E} \text{Var}(F \mid \mathcal{G}) \leq 4/N. \quad (10)$$

Second term: a smooth Gaussian function. Set

$$M(g) = D_{\phi(g)} H D_{\phi(Hg)}.$$

This is a contraction because $|\phi| \leq 1$, and the conditional mean is $\bar{F} = \sigma u^\top M(G_1) M(G_2) M(G_3) u$. We first establish a useful symmetry of partial chains. For a binary shift s define $(P_s v)_i = v_{i+s}$ and $(C_s v)_i = (-1)^{s \cdot i} v_i$. The identities $HP_s = C_s H$, $HC_s = P_s H$, and oddness of ϕ give

$$M(P_s g) = P_s M(g), \quad M(C_s g) = M(g) P_s. \quad (11)$$

Both $P_s G$ and $C_s G$ have the same law as G . Thus any vector obtained by applying an independent chain of M 's or transposed M 's to u has translation-invariant distribution: change the Gaussian in its outermost factor using (11). Its norm is at most one. All coordinate second moments are equal, so

$$\mathbb{E}|v_i|^2 = \frac{1}{N} \mathbb{E}\|v\|^2 \leq \frac{1}{N}. \quad (12)$$

The empty chain u obeys the same bound.

At one of the three links, write the conditional mean, up to σ , as $p^\top M(G) v$. The left and right partial chains p, v and the current G are mutually independent; both chains satisfy (12). Differentiating this smooth expression and using $\|a + b\|^2 \leq 2\|a\|^2 + 2\|b\|^2$ gives

$$\|\nabla_G \bar{F}\|^2 \leq 2L_\phi^2 (\|p \odot H D_{\phi(HG)} v\|^2 + \|v \odot H D_{\phi(G)} p\|^2).$$

For the first norm, condition on G, v and average over p ; (12) gives

$$\mathbb{E}_p \|p \odot H D_{\phi(HG)} v\|^2 \leq \frac{1}{N} \|H D_{\phi(HG)} v\|^2 \leq \frac{1}{N}.$$

The other norm is symmetric. The Gaussian Poincare inequality, proved in Appendix A, now gives

$$\text{Var } \bar{F} \leq \sum_{s=1}^3 \mathbb{E} \|\nabla_{G_s} \bar{F}\|^2 \leq 12L_\phi^2 / N.$$

Combining this with (10) in (9) yields $\text{Var } F \leq (4 + 12L_\phi^2)/N < 1540/N$. $\square$

7 Closing the lower bound

The mean margin (8), Lemma 3, and Chebyshev's inequality imply, for $N \geq 2^{30}$,

$$\varepsilon_\sigma := \mu_\sigma\{\sigma F < 1/4\} \leq \frac{1540 \cdot 260^2}{N} = \frac{104\,104\,000}{N} < \frac{1}{10}. \quad (13)$$

The last step follows from $1\,041\,040\,000 < 2^{30}$. A correct acceptance probability $f \in [0, 1]$ obeys $f \geq 2/3$ on the positive promise and $f \leq 1/3$ on the negative promise. Even allowing the worst behavior off the promises,

$$\mathbb{E}_{\mu_+} f \geq \frac{2}{3}(1 - \varepsilon_+), \quad \mathbb{E}_{\mu_-} f \leq \frac{1}{3}(1 - \varepsilon_-) + \varepsilon_-.$$

Thus correctness requires

$$\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f \geq \frac{1}{3} - \frac{2}{3}(\varepsilon_+ + \varepsilon_-) > \frac{1}{5}. \quad (14)$$

Suppose instead $D < (2/15)N^{1/8}$. Then $2D < \sqrt{N}$, so all relevant moments satisfy (7). Substitute that bound and $m = 4N$ into (5), using $\binom{2D}{r} \leq (2eD/r)^r$:

$$|\mathbb{E}_{\mu_+} f - \mathbb{E}_{\mu_-} f| \leq \sum_{\substack{4 \leq r \leq 2D \\ r \text{ even}}} \left(2^{3/4} e D r^{-1/4} N^{-1/8} \right)^r.$$

For $r \geq 4$ each base is at most $\theta = 2^{1/4} e D N^{-1/8} < 1/2$, since $e < 3$ and $2^{1/4} < 5/4$. The even-level sum is therefore at most

$$\sum_{j=2}^{\infty} \theta^{2j} = \frac{\theta^4}{1 - \theta^2} < \frac{1}{12}.$$

This contradicts (14) and proves the second part of Theorem 1. The argument has not conditioned the laws: accounting directly for their small off-promise mass is sufficient.

8 The finite question at $N = 4096$

Can a parallel probe of dose six solve the same task at $N = 4096$? The asymptotic estimate is too weak to answer this. We now give a finite construction and an explicit reduction of the proposed obstruction to a matrix estimate. The reduction is proved; the matrix estimate is open. All its coefficients are defined below, so the question has no implicit numerical or documentary inputs.

8.1 An exact plant

Let $q = 64$, $N = q^2$, and identify a coordinate with a pair (a, b) of binary vectors of length six. Write $S_q = \sqrt{q} H_q$, so its entries are signs and $H_N = H_q \otimes H_q$. For a uniform permutation π of these q labels and independent uniform signs σ_b , define arrays

$$L_{a,b} = S_q(a, \pi(b))\sigma_b, \quad R_{\pi(b),a} = \sigma_b S_q(b, a). \quad (15)$$

They obey $H_N L = R$ and $H_N R = L$. Indeed, with $b_c = \pi^{-1}(c)$,

$$(H_N L)_{c,d} = \frac{1}{q} \sum_{a,b} S_q(c, a) S_q(d, b) S_q(a, \pi(b)) \sigma_b = S_q(d, b_c) \sigma_{b_c} = R_{c,d};$$

the other identity follows from $H_N^2 = I$. Take three independent links and form

$$z = (L_1, R_1 \odot L_2, R_2 \odot L_3, R_3). \quad (16)$$

Starting on the right of (1), the transform identities successively cancel $R_3, L_3, R_2, L_2, R_1, L_1$ and leave u . Thus $F(z) = 1$ exactly; negating the first block gives $F = -1$.

Independently multiply every coordinate by a sign of mean $\beta = 19/25$, equivalently flipping it with probability $3/25$. Call the resulting positive law $\tilde{P}_+$ and its first-block reflection $\tilde{P}_-$. Their means are $\pm\beta^4$. Condition each law onto its respective promise to obtain P_+, P_- . For probability laws P, Q , we use $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$, half their L^1 distance.

Lemma 4 (Cost of conditioning). *Let*

$$V_\beta = \frac{(1 + \beta^2)(1 + \beta^4)}{N}, \quad g = \beta^4 - \frac{1}{4}, \quad \kappa = 2 \exp\left(-\frac{g^2}{2V_\beta}\right). \quad (17)$$

Conditioning changes the output total variation of any experiment by at most κ , and $\kappa < 1/400$.

Proof. Fix an exact plant z and reveal the four noisy blocks from left to right. The conditional expectations of F form a martingale. At block b , averaging future noise contributes β^{4-b} . The remaining deterministic right chain is a sign vector divided by $\sqrt{N}$, by the plant identities, and the left chain has norm one. Thus the martingale increment is a sum of independent centered signs with sum of squared coefficients $\beta^{2(4-b)}/N$.

A centered variable in an interval of length two satisfies $\mathbb{E}e^{tY} \leq e^{t^2/2}$: for its log moment-generating function, the second derivative is a tilted variance at most one, and its value and first derivative vanish at zero. Apply this conditionally to each sign and then each block. The four increments give $\mathbb{E}e^{t(F-\beta^4)} \leq e^{t^2V_\beta/2}$. Markov's inequality, optimized at $t = -g/V_\beta$, bounds the promise failure by $\exp(-g^2/(2V_\beta))$. Reflection gives the same bound on the negative side.

Conditioning a law on an event of probability $1 - \epsilon$ changes it in total variation by exactly ϵ (split the law into its on-event and off-event parts). An experiment cannot increase total variation. The triangle inequality gives the sum of the two failures, which is (17). For an elementary numerical bound, direct rational substitution gives

$$\frac{g^2}{2V_\beta} = \frac{546\,296\,776\,992}{80\,258\,243\,125} > \frac{34}{5}.$$

Also $e > 27/10$ from its positive power series, and $e^{4/5} > 1 + 4/5 + (4/5)^2/2 = 53/25$. Hence $e^{34/5} > (27/10)^6(53/25) > 800$, proving $\kappa < 1/400$. $\square$

8.2 The exact matrix seen by a parallel probe

For a Fock occupation ν , define its parity support in each block by $S_b(\nu) = \{i : \nu_{b,i} \text{ is odd}\}$. The mask acts only through this support: $U_x|\nu\rangle = \chi_{S(\nu)}(x)|\nu\rangle$. For hard dose six, $|S| := \sum_b |S_b| \leq 6$. Purify any input and group all terms with support S into a vector ψ_S , including their idlers. These vectors are mutually orthogonal, because they contain disjoint collections of occupation-basis vectors. Set $p_S = \|\psi_S\|^2$.

For the unconditioned laws define

$$K_{ST} = \frac{1}{2}(\mathbb{E}_{\tilde{P}_+} - \mathbb{E}_{\tilde{P}_-})\chi_{S\Delta T}. \quad (18)$$

The map $|S\rangle \mapsto \psi_S/\sqrt{p_S}$ is an isometry for nonzero weights. Therefore half the difference of the two averaged purified output states has trace norm

$$\|\text{diag}(\sqrt{p})K\text{diag}(\sqrt{p})\|_1. \quad (19)$$

The trace norm is the sum of singular values. For a trace-zero Hermitian matrix X , its positive and negative parts have equal trace, giving $\sup_{0 \leq E \leq I} |\text{tr}(EX)| = \|X\|_1/2$. Thus (19) bounds every measurement's output total variation, even after a purifier is discarded. Mixtures are covered by purification as well.

Globally negating both halves (L_j, R_j) of one link preserves its law. Consequently a plant moment vanishes unless adjacent block degrees have the same parity. First-block reflection in (18) selects the odd case: only differences with an odd number of labels in *every* block survive. Independent noise multiplies a degree- r moment by β^r .

8.3 From mode labels to 210 count states

Here is a completely specified finite criterion. For nonnegative integer four-vectors s, t with $|s|, |t| \leq 6$, let A_b, B_b range over sets of sizes s_b, t_b . Define the complete kernel and its coefficient by

$$\mathcal{O}_{s,t}(A, B) = \mathbf{1}_{\{A_b \cap B_b = \emptyset \text{ for all } b\}} \mathbb{E}_z \chi_{A \cup B}(z), \quad (20)$$

$$c_{s,t} = \sup_{\alpha, \omega} \|\text{diag}(\sqrt{\alpha}) \mathcal{O}_{s,t} \text{diag}(\sqrt{\omega})\|_1. \quad (21)$$

The expectation is the finite uniform average over the three independent permutations and sign families in (15); thus it is explicitly defined. The supremum is over all probability laws on the row and column sets, including correlated laws between blocks. These are finite matrices and compact probability simplexes, so the suprema exist. Transposition gives $c_{s,t} = c_{t,s}$. The disjointness indicator is part of the matrix, not a removable qualification on selected entries.

Write $|n| = \sum_b n_b$ and $\binom{n}{r} = \prod_{b=1}^4 \binom{n_b}{r_b}$. Index a symmetric nonnegative matrix B by

$$\mathcal{N} = \{n \in \mathbb{Z}_{\geq 0}^4 : |n| \leq 6\}, \quad |\mathcal{N}| = \binom{10}{4} = 210.$$

For $n, m \in \mathcal{N}$, set

$$B_{nm} = \mathbf{1}_{\{n_b + m_b \text{ odd } \forall b\}} \sum_{0 \leq r \leq \min(n, m)} \beta^{|n|+|m|-2|r|} c_{n-r, m-r} \sqrt{\binom{n}{r} \binom{m}{r}}. \quad (22)$$

Inequalities between four-vectors are componentwise.

Proposition 5 (Finite spectral reduction). *For every parallel probe of hard dose at most six, including arbitrary idlers, repeated occupations and photon-number coherence, the output laws under P_+, P_- satisfy*

$$\text{TV}(T_+, T_-) \leq \lambda_{\max}(B) + \kappa. \quad (23)$$

In particular, $\lambda_{\max}(B) + \kappa < 1/3$ would rule out a worst-case error of $1/3$ at dose six.

Proof. First fix the block counts $n_b = |S_b|$, $m_b = |T_b|$ and the common support $R = S \cap T$, with counts $r_b = |R_b|$. Write $S = R \dot{\cup} A$, $T = R \dot{\cup} B'$. The exclusive counts are $s = n - r$, $t = m - r$. The corresponding block of K is the kernel (20), restricted to sets avoiding R , multiplied by $\beta^{|s|+|t|}$. It vanishes unless $n_b + m_b$ is odd for every block.

To bound this block, temporarily take normalized laws p_S, w_T on the two count sectors. For a fixed R put $P_R = \sum_A p_{R \cup A}$ and $W_R = \sum_{B'} w_{R \cup B'}$. The definition (21), applied to the restricted normalized laws, bounds the weighted trace norm by $c_{s,t} \sqrt{P_R W_R}$ before the noise factor. Each nonzero entry has the unique residual $R = S \cap T$, so summing over all residuals reconstructs the desired block exactly. Triangle inequality and Cauchy-Schwarz give

$$\sum_R c_{s,t} \sqrt{P_R W_R} \leq c_{s,t} \sqrt{\binom{n}{r} \binom{m}{r}}.$$

Indeed each S contains exactly $\binom{n}{r}$ eligible residual sets, and each T exactly $\binom{m}{r}$.

Now let $v_n = \sqrt{\sum_{S: |S_b|=n_b} p_S}$ for the actual input. Restoring sector masses supplies a factor $v_n v_m$. Sum over common counts r and ordered sectors n, m , using triangle inequality again. Equations (19) and (22) give

$$\text{TV}(\tilde{T}_+, \tilde{T}_-) \leq v^\top B v \leq \lambda_{\max}(B), \quad \|v\|_2 = 1.$$

Lemma 4 adds at most κ . Finally, equal-prior optimal decision error is $(1 - \text{TV})/2$: choose the more likely law for each output and sum the pointwise minima. A worst-case error at most $1/3$ would give equal-prior error at most $1/3$, requiring $\text{TV} \geq 1/3$. $\square$

8.4 A worked example: dose two

The criterion does give a concrete finite conclusion at smaller dose. With hard dose two, any surviving difference uses all four blocks, so the two supports must be disjoint, of size two each, and occupy complementary pairs of blocks. For coordinates $i = (a, b)$, $j = (c, d)$, sign averaging in (15) leaves only $\pi(b) = c$; hence

$$\mathbb{E}L_i R_j = \frac{1}{q} S_q(a, c) S_q(b, d) = H_N(i, j).$$

The three independent links therefore make each entry of a surviving kernel a product of three Walsh entries, of magnitude $N^{-3/2}$. Each row has N^2 columns, so its squared norm is $1/N$. For arbitrary probability laws α, ω , factor the weighted matrix as $(\text{diag}(\sqrt{\alpha})\mathcal{O})\text{diag}(\sqrt{\omega})$. The inequality $\|XY\|_1 \leq \|X\|_F \|Y\|_F$, where the Frobenius norm is the square root of the sum of squared entries, gives $c_{s,t} \leq 1/\sqrt{N}$. This inequality follows by duality: for $\|U\| \leq 1$, apply Cauchy–Schwarz to $\text{tr}(UXY)$ and use $\|UX\|_F \leq \|X\|_F$.

Restricting the count states to $|n| \leq 2$ leaves three complementary two-state blocks and zero rows. Each edge is at most $\beta^4/\sqrt{N}$. Thus, for every parallel probe of dose at most two at $N = 4096$,

$$\text{TV}(T_+, T_-) \leq \frac{\beta^4}{64} + \kappa < \frac{192821}{25000000} < \frac{1}{100}.$$

Its equal-prior error exceeds $99/200$. This is a proved finite obstruction; the dose-six problem requires higher-degree kernel bounds.

8.5 What remains at dose six

The entries of B are defined entirely by finite averages and finite norm optimizations. No bound $\lambda_{\max}(B) + \kappa < 1/3$ is proved here. Establishing it requires estimates for the *complete* kernels (20); accurate numerical arithmetic on proposed upper bounds cannot supply a missing analytic inequality. This matrix estimate is sufficient, not necessary: failure to prove it, or even its failure, would not exclude a sharper lower-bound argument.

One sometimes imposes an extra restriction that makes this calculation smaller. Define the parity-size operator

$$Q|\nu\rangle = |S(\nu)| |\nu\rangle.$$

Extend Q by the identity on idlers. For an input density operator ϱ with $[\varrho, Q] = 0$, decompose into fixed- Q sectors; only pairs with $|n| = |m|$ are needed. In (22) this means replacing B by $B_{nm}^{\text{bal}} = \mathbf{1}_{\{|n|=|m|\}} B_{nm}$. The same proof applies sector by sector and then by convexity. Even the corresponding smaller matrix estimate remains open.

Fixed physical photon number does not justify this restriction. For example, three photons in coordinate zero of block one have parity size one. One photon in coordinate zero of each of blocks two, three and four has parity size three. Their superposition has fixed physical number three but is not diagonal in Q . Its connecting moment is nonzero: each link has $\mathbb{E}L_{0,0}R_{0,0} = 1/q$, because sign averaging leaves precisely $\pi(0) = 0$. Thus (18) for these two supports equals $\beta^4/q^3 > 0$. Deleting it would change the experiment. Appendix B gives another explicit obstruction to a shortcut in estimating the finite coefficients.

9 The remaining adaptive question

The proved lower bound concerns a fixed parallel input and measurement. If later probes depend on earlier outcomes, the common fixed operator used in (2) is no longer available. For fresh batches with classical feedback and total hard dose D , the acceptance probability still has degree at most $2D$: along a fixed history each unnormalized amplitude contains at most D mask-sign factors. Its probability has degree at most $2D$, and summing histories preserves this bound. Degree alone is insufficient. A sufficient additional estimate would be

$$\left| \sum_{|S|=r} c_S \hat{f}(S) \right| \leq \binom{2D}{r} (4N)^{r/4} \max |c_S|.$$

This is twice the parallel bound. It would give a gap below $1/6$ in Section 7, still contradicting the required gap above $1/5$. We have not established this estimate for adaptive preparations. Neither a mean-dose bound nor optimality of the dose-six protocol is claimed.

A The probability tools used in the proof

The following short derivations make the Gaussian and concentration steps independent of external references.

A.1 The Gaussian sign identity

If standard Gaussians X, Y have correlation c , write $X = G_1$, $Y = cG_1 + \sqrt{1-c^2} G_2$ with independent standard G_1, G_2 . The angle in this plane is uniform. The two signs disagree on wedges of total angle $2 \arccos c$, so

$$\Pr(\operatorname{sgn} X \neq \operatorname{sgn} Y) = \frac{\arccos c}{\pi}, \quad \mathbb{E} \operatorname{sgn} X \operatorname{sgn} Y = \frac{2}{\pi} \arcsin c.$$

The endpoint correlations follow by continuity.

A.2 Hermite degrees and Gaussian contraction

Define the normalized Hermite polynomials by

$$e^{tx-t^2/2} = \sum_{n \geq 0} h_n(x) \frac{t^n}{\sqrt{n!}}.$$

Taking Gaussian expectations of two generating functions shows $\mathbb{E} h_n(G) h_m(G) = \mathbf{1}_{\{n=m\}}$. The products $h_\alpha(x) = \prod_i h_{\alpha_i}(x_i)$ form an orthonormal basis in Gaussian L^2 . One way to see completeness is to take a function orthogonal to all polynomials. Its Gaussian-weighted Laplace transform has all derivatives zero at zero and is entire by Cauchy-Schwarz and Gaussian exponential integrability. It therefore vanishes also at imaginary arguments; uniqueness of the Fourier transform makes the original function zero.

The total degree of h_α is $|\alpha| = \sum_i \alpha_i$. The degree- p space is invariant under orthogonal rotations, since it is the orthogonal complement of degrees at most $p-1$ inside the polynomials of degree at most p . If standard Gaussian vectors X, Y have cross-covariance R , their generating functions satisfy

$$\mathbb{E} e^{t \cdot X - \|t\|^2/2} e^{s \cdot Y - \|s\|^2/2} = e^{t^\top R s}.$$

Rotate X, Y by a singular-value decomposition of R (padding to the same dimension with independent coordinates if necessary). If its singular values are s_j , expansion in these rotated coordinates gives

$$\mathbb{E}h_\alpha(X)h_\gamma(Y) = \mathbf{1}_{\{\alpha=\gamma\}} \prod_j s_j^{\alpha_j}.$$

Thus different total degrees do not pair, and degree p contracts by at most $\|R\|^p$. If f, g have no degrees below a, b , then, writing f_p, g_p for their degree- p parts and using $\|R\| \leq 1$,

$$|\mathbb{E}f(X)g(Y)| \leq \sum_{p \geq \max(a,b)} \|R\|^p \|f_p\|_2 \|g_p\|_2 \leq \|R\|^{\max(a,b)} \|f\|_2 \|g\|_2.$$

A product of signs in a distinct coordinates is odd in every one of them. Each participating Hermite index is therefore odd, giving minimum degree a and L^2 norm one, as used in (6).

A.3 Gaussian Poincare inequality

Let $f = \sum_\alpha c_\alpha h_\alpha$ be smooth with square-integrable gradient. Gaussian integration by parts gives

$$\langle \partial_i f, h_\gamma \rangle = \sqrt{\gamma_i + 1} c_{\gamma + e_i}.$$

Parseval's identity therefore yields

$$\text{Var } f = \sum_{\alpha \neq 0} |c_\alpha|^2 \leq \sum_\alpha |\alpha| |c_\alpha|^2 = \mathbb{E} \|\nabla f\|^2.$$

The smooth functions used in the proof have bounded derivatives, so the integrations are justified by Gaussian integrability.

A.4 Variance tensorization for independent variables

For independent $Y_1, \dots, Y_k$, let E_i average coordinate i alone and let $F_i = \mathbb{E}[F \mid Y_1, \dots, Y_i]$. The martingale differences are orthogonal, and independence gives

$$F_i - F_{i-1} = \mathbb{E}[F - E_i F \mid Y_1, \dots, Y_i].$$

Conditional Jensen and summation imply

$$\text{Var } F \leq \sum_i \mathbb{E}(F - E_i F)^2 = \sum_i \mathbb{E} \text{Var}_{Y_i}(F \mid Y_{-i}).$$

If Y_i is a sign and $F = A + Y_i B$ with A, B independent of that coordinate, the last conditional variance is $(1 - (\mathbb{E}Y_i)^2)B^2 \leq B^2$. This proves the coefficient bound used in (10).

B Why a finite coefficient shortcut fails

This example explains why the finite matrix estimate is a substantive remaining task. For the link (15), define

$$M(A, B) = \mathbb{E} \prod_{v \in A} L_v \prod_{w \in B} R_w.$$

For any power of two $q \geq 8$, use the three-cell sets

$$V = \{(0, 0), (1, 0), (2, 0)\}, \quad W = \{(0, 0), (0, 1), (0, 2)\}.$$

Here 0, 1, 2 denote their distinct binary labels. For $M(V, W)$, the sign in column zero must match the sign in column $\pi^{-1}(0)$, so averaging leaves exactly $\pi(0) = 0$. Every remaining Walsh phase is +1, giving $M(V, W) = 1/q$. For $M(W, V)$, sign averaging requires the permutation to map $\{0, 1, 2\}$ onto itself; again all surviving phases are +1. Thus

$$M(V, W) = \frac{1}{q}, \quad M(W, V) = \frac{1}{\binom{q}{3}}.$$

The exact four-block plant moment at (V, W, V, W) is consequently $1/(q^2 \binom{q}{3})$. In particular, a proposed universal maximum of the form

$$\frac{b^2}{\binom{q}{3}}, \quad b = \frac{q+2}{q(q-1)(q-2)},$$

is false: $b < 1/q$ for $q \geq 8$. At $q = 8$ the actual entry is $1/3584$, exceeding the proposed $25/1580544$. At $q = 64$ the same comparison applies. This refutes that entrywise shortcut; it does not refute a possible bound on the full weighted trace norms (21). Those require a valid argument that retains the whole kernel.